FUNNOW Group — AI Management Trainee Technical Assignment

1. Your Role

Act as my **Senior AI Automation Engineer, Software Architect, and Technical Mentor**.

Help me design, implement, test, document, and prepare a submission-ready solution for FUNNOW Group's AI Management Trainee assignment.

I want a genuinely working implementation, not just pseudocode, a conceptual architecture, or a presentation.

Work collaboratively with me and explain important implementation decisions. I should be able to understand the system, run it myself, troubleshoot problems, and confidently explain it during a technical interview.

2. My Background

I am a fourth-year Software Engineering student at Mae Fah Luang University, Thailand, with a GPA of 3.88.

My career interests include Data Science, Machine Learning, AI, Data Engineering, and AI-powered business applications.

Project 1: AgriCredit Resilience

An explainable machine-learning decision-support system developed through the Myanmar Machine Learning Bootcamp under the Frontier Tech Leaders Programme.

My experience includes:

- Python, pandas, NumPy, scikit-learn
- Data cleaning, preprocessing, feature engineering
- Classification using Decision Tree and Logistic Regression
- Model training, evaluation, and comparison
- Metrics: accuracy, precision, recall, F1-score, ROC-AUC
- Streamlit dashboard development
- Translating technical predictions into understandable decision-support insights

Project 2: SoM-BI

A team-based senior project developing a Business Intelligence platform for the School of Management at Mae Fah Luang University.

My experience includes:

- SQL and MySQL
- Node.js and Express.js
- REST API integration
- Data validation and integration
- KPI reporting and dashboards
- Nuxt 3, Vue 3, Firebase, Chart.js
- Git, GitHub, Docker, and basic cloud concepts

My Technical Level

I have a good foundation in programming, machine learning, data processing, databases, and API-based applications.

However, I am still learning:

- n8n workflow automation
- Production AI agents
- LLM API integration
- Prompt evaluation
- Production workflow monitoring

Do not assume I already know these technologies. Explain unfamiliar components and why they are necessary.

Prefer technologies that allow me to leverage my existing Python and JavaScript knowledge.

3. The Official FUNNOW Assignment

Business Situation

Every Monday morning, FUNNOW's hotel partners want a summary of their performance during the previous week.

Currently, employees manually retrieve data, analyze hotel performance, and write individual summaries. The process is time-consuming and error-prone.

Required System

Build an automated workflow that:

1. Retrieves hotel performance data from the provided Google Sheet.
2. Identifies hotels performing well versus those needing improvement.
3. Generates a personalized performance summary for each hotel.
4. Sends the summary to the hotel owner using email.
5. Uses my own email address for delivery testing.

FUNNOW explicitly allows any programming language or tool.

Official Deliverables

1. A working version of the automated system.
2. A short write-up explaining:
    - Why the technologies were selected.
    - How the performance classification thresholds were determined.
    - How AI was designed to generate hotel-specific messages.

4. Actual Google Sheet

Source:

https://docs.google.com/spreadsheets/d/18mr5ZyoyYDuuJ0TB4hncKw2CWUxBAjtIupadEoVNz_c/edit

The spreadsheet was previously inspected and contains two tabs:

Sheet1: Mock hotel performance data.

Columns:

- `brandname` — Hotel name
- `revenue` — Revenue
- `Order` — Booking/order count
- `Boss Gmail` — Owner email address

There are 10 hotel records.

Dispatch Log: An existing log containing previous testing records.

Important:

- The mock data does not contain a date column.
- It does not contain revenue or booking targets.
- It does not provide previous-week comparisons.

- The revenue currency is not explicitly specified.
- There is an existing dispatch log which may contain work from other tests.

Do not modify the original spreadsheet without my explicit permission.

Do not assume that a particular record represents a verified calendar week. Provide a clearly labeled reporting-period assumption or configurable reporting period for the demonstration.

For production readiness, explain how dated records, business targets, and historical comparisons would be added.

5. Recommended Technical Architecture

My initial preference is:

Google Sheets → n8n → Data Validation and Analysis → LLM → Email → Dispatch Log

I am open to using Python for data processing, classification, and validation.

You may propose a better architecture if it improves reliability, simplicity, or suitability for FUNNOW's job requirements.

The solution should demonstrate business automation skills, not unnecessary architectural complexity.

Main Components

A. Workflow Trigger

- Manual trigger for development and testing.
- Scheduled Monday trigger for production demonstration.
- Clear reporting-period handling.
- Use an explicit time zone, preferably Asia/Bangkok.

B. Google Sheets Integration

- Read hotel records.
- Validate required fields.
- Convert revenue and order counts into numeric values.
- Handle missing, malformed, or inconsistent data.
- Prevent duplicate hotel processing.

C. Performance Classification

Implement transparent and explainable classification rules using deterministic code.

Possible starting approach:

- Exclude invalid records and special zero-data records from benchmark calculation.
- Calculate median revenue and median orders among valid hotels.
- Classify hotels meeting both median benchmarks as Doing Well.
- Classify hotels below either benchmark as Needs Improvement.
- Flag zero-revenue and zero-order records for Data Check instead of automatically treating them as poor performance.

These thresholds are suggestions, not fixed business requirements.

Evaluate whether they make sense, propose improvements if justified, and document the reasoning.

Previously observed positive-record medians were approximately:

- Revenue: 315,000
- Orders: 71

Recalculate them from the live dataset rather than hardcoding these values.

Do not invent performance goals, percentage growth, occupancy rates, or historical trends that the sheet does not contain.

D. Personalized AI Summary Generation

Use an LLM to generate professional, friendly, data-grounded messages.

The AI should receive structured inputs such as:

- Hotel name
- Revenue
- Order count
- Performance classification
- Benchmark comparisons
- Reporting period
- Suggested business considerations

The AI must:

- Use only provided facts.
- Avoid invented statistics.
- Never claim that performance improved or declined without historical data.
- Adapt the message according to performance.
- Offer constructive recommendations.
- Maintain a professional tone.

- Avoid revealing other hotels' identifiable business data.
- Return structured, validated output where practical.

Calculations and classifications must happen in code, not be delegated to the language model.

Implement a clearly identified deterministic fallback message if the LLM API fails.

Do not falsely label fallback-generated content as AI-generated.

E. Email Delivery

Generate an appropriate email subject and formatted body for each hotel.

For testing:

- Send only to my explicitly configured test email.
- Never send messages to the hotel addresses listed in the mock data during testing.
- Separate original owner addresses from actual test recipients.
- Require explicit configuration before enabling live partner delivery.
- Provide a preview or dry-run mode.
- Track successful and failed deliveries.

F. Logging and Reliability

Include:

- Processing timestamp
- Reporting period
- Hotel identifier
- Classification
- AI-generation status
- Delivery status
- Error details
- Actual test recipient
- Duplicate-send prevention

Use a separate local log or my own Google Sheet for testing. Do not overwrite the original shared Dispatch Log.

Use a unique hotel-and-reporting-period key to prevent duplicate weekly emails.

Record SENT status only after confirmed delivery-provider acceptance, distinguishing that from actual inbox receipt.

Keep credentials and API keys outside source code using secure configuration.

6. Implementation Standards

Prioritize a simple, maintainable, demonstrable system.

I want:

- Clean, modular code or well-organized n8n nodes
- Understandable naming conventions
- Comments where necessary
- Clear instructions for configuring credentials
- Reliable data validation
- Safe email testing
- Error handling and fallback behavior
- Reproducible test runs
- Minimal unnecessary dependencies

Do not introduce complicated frameworks unless they add a meaningful benefit.

7. Testing Requirements

Create and execute a test plan covering:

1. Valid hotel record processing.
2. Correct performance classification.
3. Personalized summaries for different hotels.
4. Zero revenue and zero orders.
5. Invalid or missing fields.
6. Missing or invalid email addresses.
7. LLM API failure.
8. Email delivery failure.
9. Duplicate execution for the same reporting period.
10. Successful delivery to my test inbox.

Verify actual results rather than assuming the workflow succeeded.

Any functionality not tested must be clearly identified.

8. Submission Deliverables

Help me prepare:

Deliverable A — Working System

- Source code and/or exportable n8n workflow

- Configuration instructions
- Dependency information
- Safe test settings
- Runnable demonstration
- Sample generated hotel summaries
- Evidence of successful test-email delivery

Deliverable B — Short Technical Write-Up

Explain:

1. Problem and objective.
2. Architecture overview.
3. Technology-selection rationale.
4. Data preparation and validation.
5. Classification threshold methodology.
6. AI prompt design and factual safeguards.
7. Email automation.
8. Testing, limitations, and future improvements.

Use professional but simple English suitable for an internship technical assessment.

Optional: Help me prepare a short demonstration video or walkthrough showing the complete workflow.

9. How I Want You to Work With Me

Follow these stages:

Stage 1 — Requirements and Architecture

Analyze the assignment, inspect the source data, identify limitations, and recommend a practical architecture.

Stage 2 — Environment Setup

Guide me through installing and configuring the tools, including n8n if selected.

Stage 3 — Core Implementation

Build data retrieval, validation, performance classification, and message preparation.

Stage 4 — AI Integration

Implement personalized LLM-generated summaries with factual safeguards and error handling.

Stage 5 — Email and Scheduling

Implement safe test-email delivery, Monday scheduling, and dispatch logging.

Stage 6 — Testing

Run realistic tests and fix identified problems.

Stage 7 — Submission Preparation

Prepare the final implementation, documentation, and a demonstration plan.

Explain each component before or while implementing it so I understand the code and workflow.

Do not fabricate API calls, test results, successful emails, or tool execution.

10. Final Objective

I want a submission that demonstrates:

- Practical AI automation
- Data processing and analytical thinking
- Responsible LLM integration
- Business workflow improvement
- Reliable system design
- Clear technical communication

The finished solution should be technically credible, easy to demonstrate, and directly relevant to FUNNOW's AI Management Trainee responsibilities.

Start by analyzing the assignment and the linked Google Sheet. Then recommend the most appropriate architecture based on my background before beginning implementation.